



Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Friday, 7 August 2026, 7:57 PM**State** Finished**Completed on** Friday, 7 August 2026, 8:01 PM**Time taken** 4 mins 16 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int firstZero(int arr[], int low, int high) {
4      if (low > high)
5          return -1;
6
7      int mid = low + (high - low) / 2;
8
9      if (arr[mid] == 0 && (mid == 0 || arr[mid - 1] == 1))
10         return mid;
11
12     if (arr[mid] == 1)
13         return firstZero(arr, mid + 1, high);
14     else
15         return firstZero(arr, low, mid - 1);
16 }
17
18 int main() {
19     int n;
20     scanf("%d", &n);
21
22     int arr[n];
23     for (int i = 0; i < n; i++)
24         scanf("%d", &arr[i]);
25
26     int index = firstZero(arr, 0, n - 1);
27
28     if (index == -1)
29         printf("0");
30     else
31         printf("%d", n - index);
32
33     return 0;
34 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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J2

Started on	Thursday, 27 August 2026, 10:38 AM
State	Finished
Completed on	Thursday, 27 August 2026, 10:39 AM
Time taken	1 min 37 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countOccurrences(int nums[], int left, int right, int value) {
4      int count = 0;
5
6      for (int i = left; i <= right; i++) {
7          if (nums[i] == value)
8              count++;
9      }
10
11     return count;
12 }
13
14 int majorityElement(int nums[], int left, int right) {
15
16     if (left == right)
17         return nums[left];
18
19
20     int mid = left + (right - left) / 2;
21
22
23     int leftMajority = majorityElement(nums, left, mid);
24     int rightMajority = majorityElement(nums, mid + 1, right);
25
26
27     if (leftMajority == rightMajority)
28         return leftMajority;
29
30
31     int leftCount = countOccurrences(nums, left, right, leftMajority);
32     int rightCount = countOccurrences(nums, left, right, rightMajority);
33
34     if (leftCount > rightCount)
35         return leftMajority;
36     else
37         return rightMajority;

```

```
37         return majority,
38     }
39
40     int main() {
41         int n;
42         scanf("%d", &n);
43
44         int nums[n];
45
46         for (int i = 0; i < n; i++)
47             scanf("%d", &nums[i]);
48
49         int result = majorityElement(nums, 0, n - 1);
50
51         printf("%d\n", result);
52     }
```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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J2

Started on Thursday, 27 August 2026, 10:41 AM**State** Finished**Completed on** Thursday, 27 August 2026, 10:43 AM**Time taken** 2 mins 22 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findFloor(int arr[], int low, int high, int x) {
4      if (low > high) {
5          return -1;
6      }
7
8      if (x >= arr[high]) {
9          return arr[high];
10     }
11
12     int mid = low + (high - low) / 2;
13
14     if (arr[mid] == x) {
15         return arr[mid];
16     }
17
18     if (arr[mid] > x) {
19         return findFloor(arr, low, mid - 1, x);
20     }
21
22     int floorInRight = findFloor(arr, mid + 1, high, x);
23     return (floorInRight != -1) ? floorInRight : arr[mid];
24 }
25
26 int main() {
27     int n, x;
28
29     if (scanf("%d", &n) != 1) return 0;
30
31     int arr[n];
32     for (int i = 0; i < n; i++) {
33         scanf("%d", &arr[i]);
34     }
35
36     scanf("%d", &x);
37
38     int floorValue = findFloor(arr, 0, n - 1, x);
39
40     printf("%d\n", floorValue);
41
42     return 0;
43 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 27 August 2026, 11:24 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:31 AM
Time taken	7 mins 2 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3
4  int recursive_binary_search(int arr[], int low, int high, int target) {
5      if (low > high) {
6          return 0;
7      }
8
9      int mid = low + (high - low) / 2;
10
11     if (arr[mid] == target) {
12         return 1;
13     } else if (arr[mid] > target) {
14         return recursive_binary_search(arr, low, mid - 1, target);
15     } else {
16         return recursive_binary_search(arr, mid + 1, high, target);
17     }
18 }
19
20 int find_pair_recursive(int arr[], int low, int high, int x, int *elem1, int *elem2) {
21     if (low >= high) {
22         return 0;
23     }
24
25     int current_element = arr[low];
26     int complement = x - current_element;
27
28     if (recursive_binary_search(arr, low + 1, high, complement)) {
29         *elem1 = current_element;
30         *elem2 = complement;
31         return 1;
32     }
33
34     return find_pair_recursive(arr, low + 1, high, x, elem1, elem2);
35 }
36
37 int main() {
38     int n;
39     if (scanf("%d", &n) != 1) return 0;
40
41     int arr[n];
42     for (int i = 0; i < n; i++) {
43         scanf("%d", &arr[i]);
44     }
45
46     int x;
47     scanf("%d", &x);
48
49     int elem1, elem2;
50     int found = find_pair_recursive(arr, 0, n - 1, x, &elem1, &elem2);
51
52     if (found) {

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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J2

Started on Thursday, 27 August 2026, 11:33 AM**State** Finished**Completed on** Thursday, 3 September 2026, 8:25 PM**Time taken** 7 days 8 hours**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  void quickSort(int a[], int low, int high)
4  {
5      if (low < high)
6      {
7          int i = low;
8          int j = high;
9          int pivot = a[low];
10         int temp;
11
12         while (i < j)
13         {
14             while (a[i] <= pivot && i < high)
15                 i++;
16
17             while (a[j] > pivot)
18                 j--;
19
20             if (i < j)
21             {
22                 temp = a[i];
23                 a[i] = a[j];
24                 a[j] = temp;
25             }
26         }
27
28         // Place pivot in its correct position
29         temp = a[low];
30         a[low] = a[j];
31         a[j] = temp;
32
33         // Divide and conquer
34         quickSort(a, low, j - 1);
35         quickSort(a, j + 1, high);
36     }
37 }
38
39 int main()
40 {
41     int n, i;
42     int a[100];
43
44     scanf("%d", &n);
45
46     for (i = 0; i < n; i++)
47         scanf("%d", &a[i]);
48
49     quickSort(a, 0, n - 1);
50
51     for (i = 0; i < n; i++)
52         printf("%d ", a[i]);

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)